Disruption mitigation experiments at JET

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The views and opinions expressed herein do not necessarily reflect those of the European Commission.

Motivation

JET:

- Reduce heat loads during disruption to level sufficient to avoid melting
- Mitigate vessel forces during disruptions at high plasma current

ITER:

- Test ITER-like disruption mitigation scheme using MGI(s)
- Determine best injection conditions depending on plasma energy, etc.

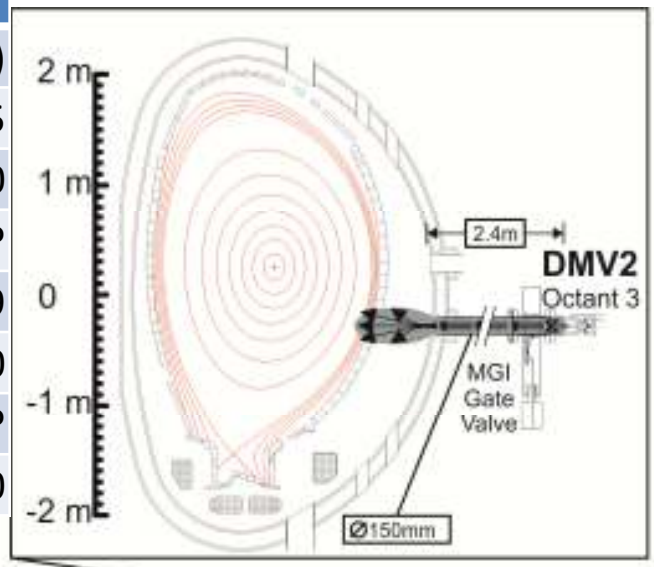
Tasks:

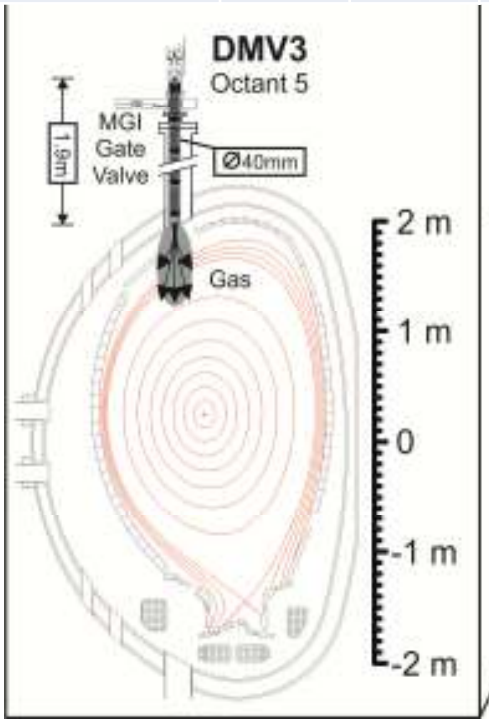
- Maximise radiated energy during disruption (ideally during TQ)
- Reduce radiation asymmetries
- Improve mixing efficiency
- Determine optimum CQ-time to minimise halo and eddy currents and vessel forces
- Improve Real-Time prediction of disruption
- Avoid appearance of runaway electrons

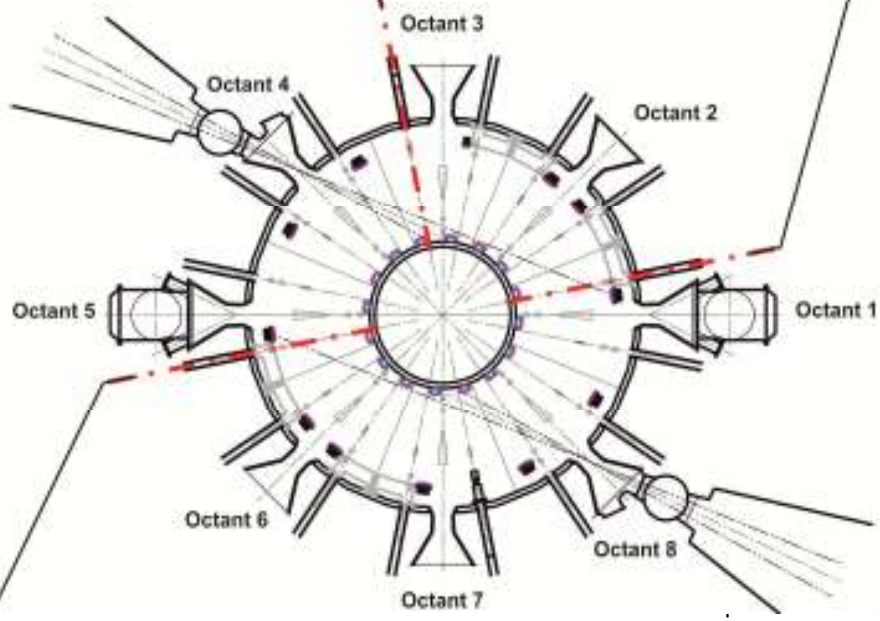
Note: disruption mitigation experiments do not have aim at disruption avoidance.

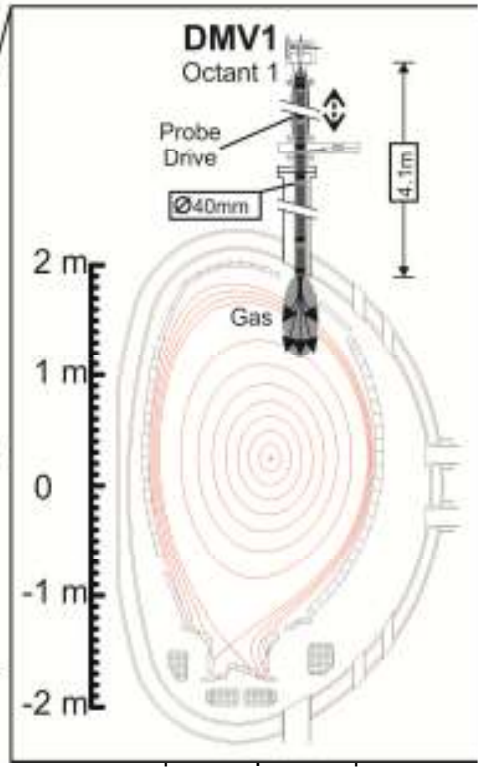
Disruption mitigation system at JET

	DMV1	DMV2	DMV3
Operational	2008	2014	(2015)
Vol [ltr]	0.65	0.975	0.35
p_{inj} [MPa]	3.6	5.0	5.0
Gas (D_2) [barL]	~10	~45	~17?
Tube [m]	4.1	2.4	1.9
Orifice [mm]	10	30	30
Trigg.delay [ms]	9	12	9-12?
ToF (D2+10%Ar) [ms]	2.0	1.2	1.0



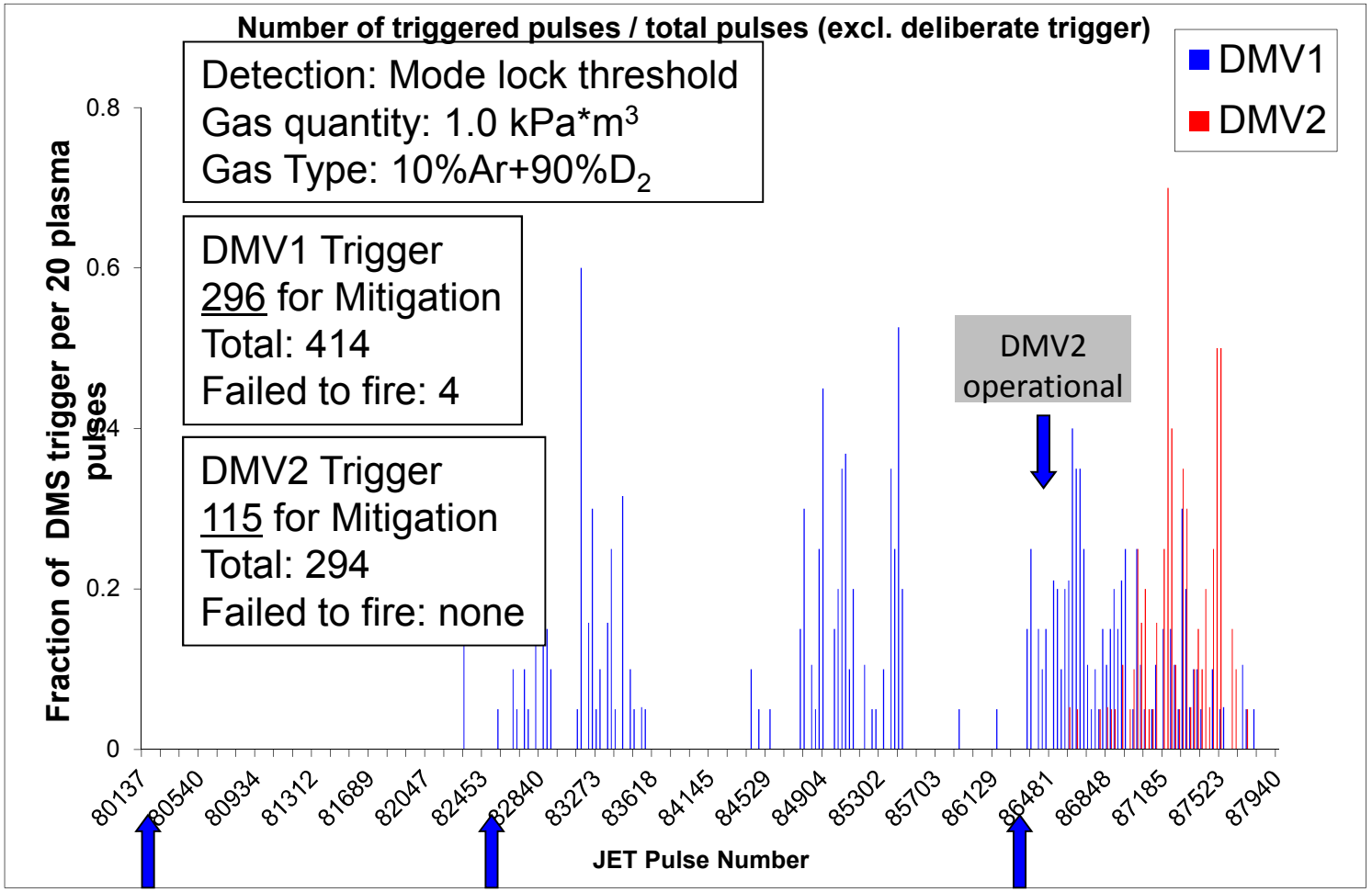






Operational experience of MGI at JET

- Routine operation has been established
- Little effect of MGI on subsequent pulses (no NSBs, benign Ar-contamination)



2011 Start of ITER-Like-Wall experiment

DMS recomended for plasmas above 2.5 MA

DMS mandatory for plasmas above 2.0 MA/5MJ

JET

Experimental strategy

MGI-experiments at JET:

- Feed-forward to test various injection settings in reproducible plasma conditions using
  - kicked VDEs or
  - plain massive gas injection in “healthy” plasmas.
- Few tests with seeded error field locked modes to study radiation asymmetries.
- Few demonstration pulses with closed-loop using mode lock-signal in density limit disruptions.
- Dual injection (DMV1+DMV2) attempted.

Parameters to optimise:

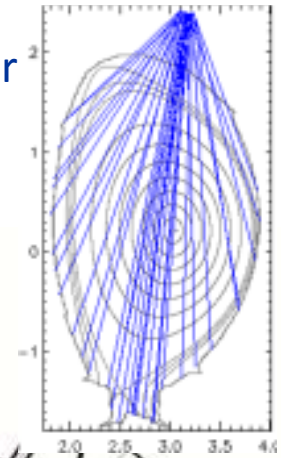
- Gas species and impurity content
- Gas amount (proportional to injection pressure)
- Injection location
- Timing of DMVs

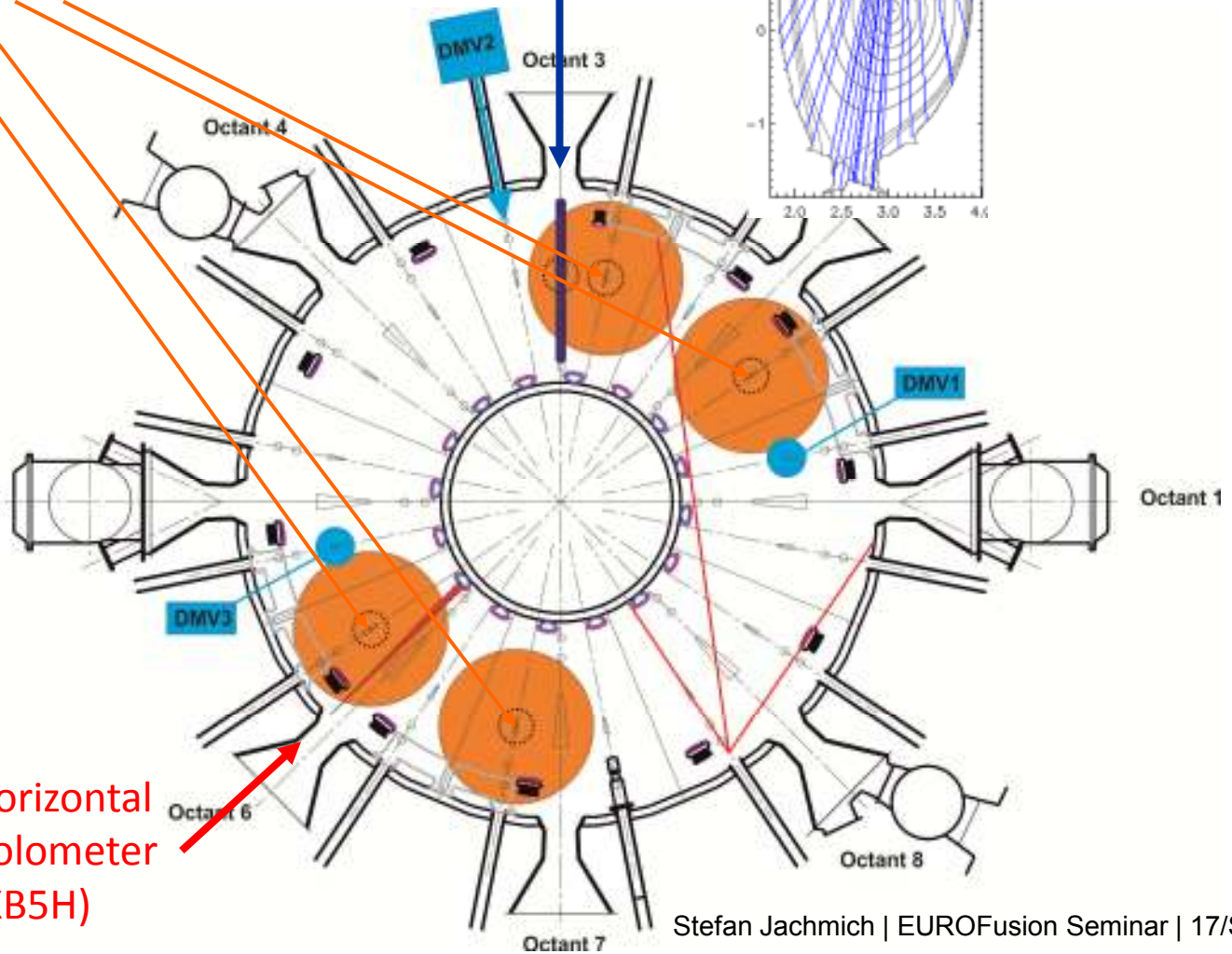
Diagnostic constraints

Pinhole Bolometer (KB1):
- 4 toroidal locations
- view from top

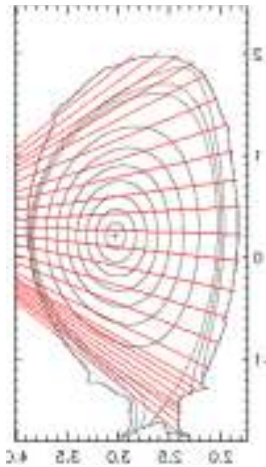
Vertical Bolometer (KB5V)

LoSs of KB5V





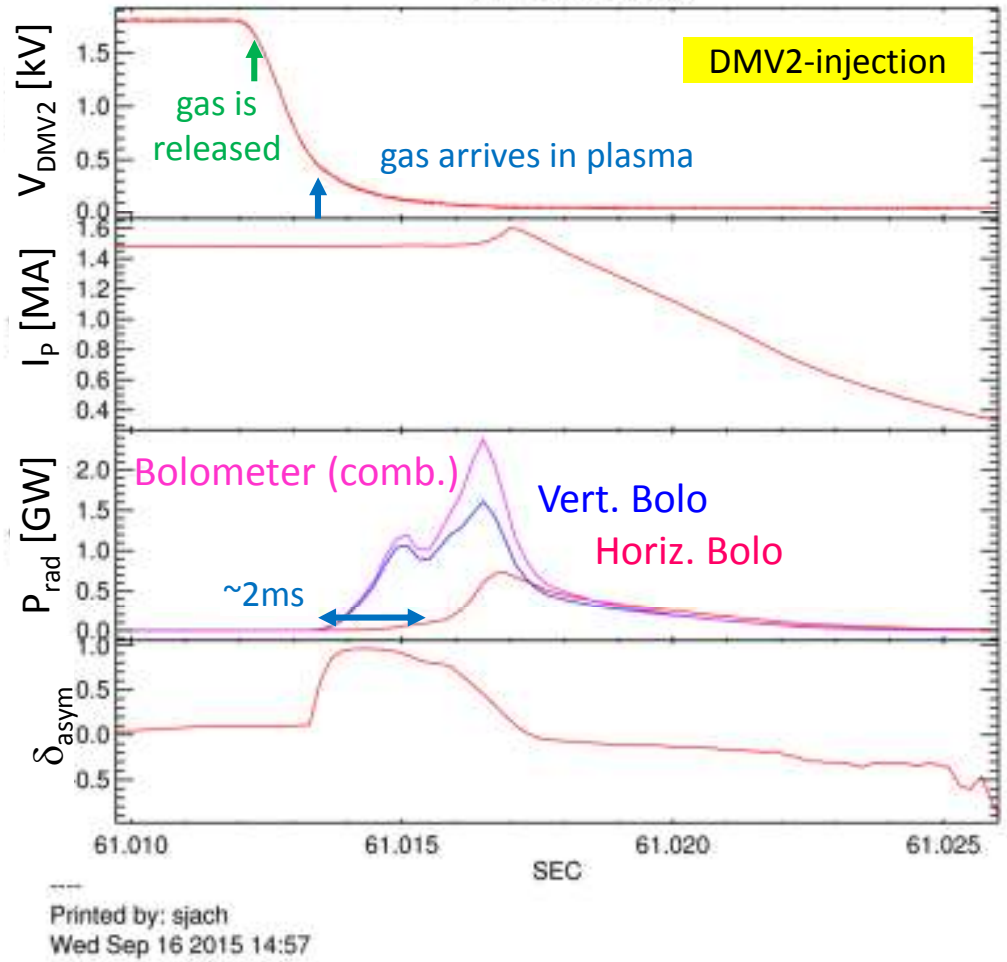
LoSs of KB5H



Horizontal Bolometer (KB5H)

DMV2-injection and vertical bolometer

- Before disruption: W_{mag} = 5.7 MJ, $W_{thermal}$ = 2.0 MJ
- Dissipates into: $W_{coupled}$ = 3.0 MJ, $W_{rad,topi}$ = 5.7 MJ and W_{cond} = –1.0MJ (f_{rad}=120%)
  - Vertical bolometer measurements are affected by the neutral gas
  - Integration of volume radiation assumes toroidal symmetry

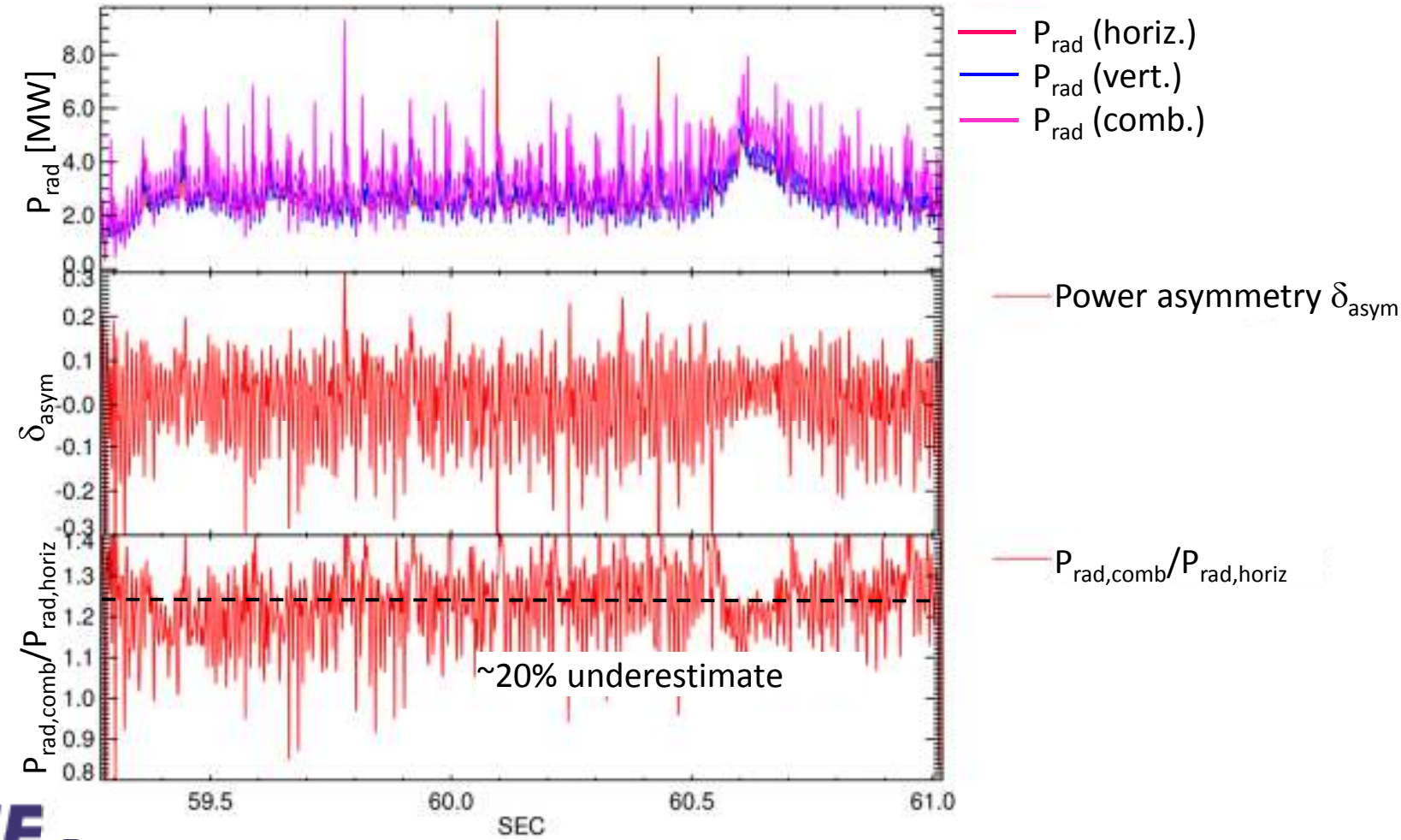


Power asymmetry δ_{asym}:

$$\frac{P_{rad,vert} - P_{rad,horiz}}{P_{rad,vert} + P_{rad,horiz}}$$

Radiation analysis: prior disruption

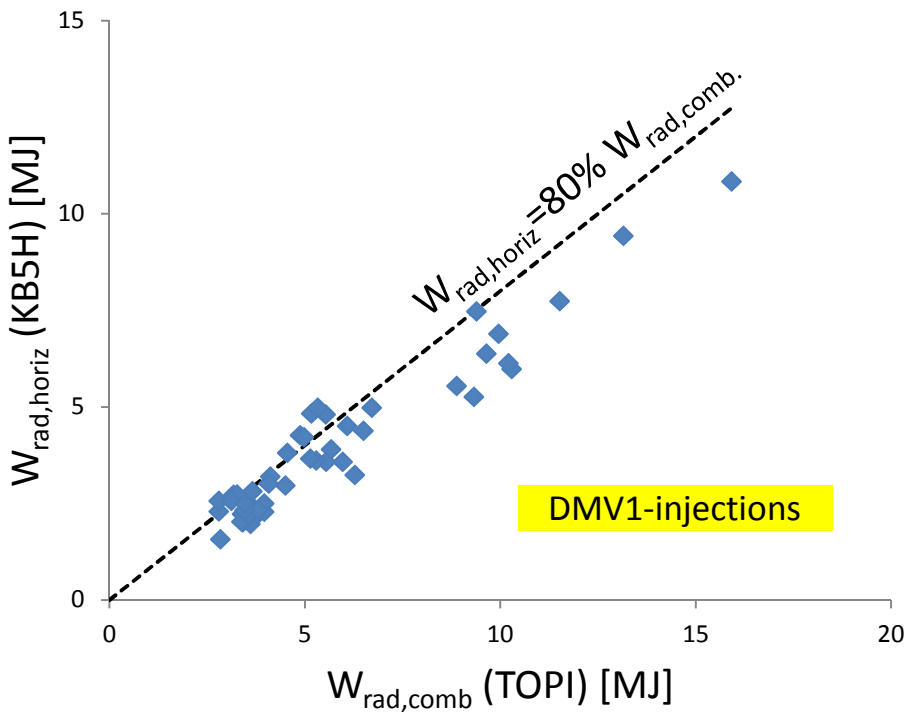
- Separate integration of channels from vertical and horizontal bolometer
- Slight asymmetry between vertical and horizontal bolometer
- Horizontal bolometer underestimates radiation by ~20%



Radiation analysis: DMV1-disruptions

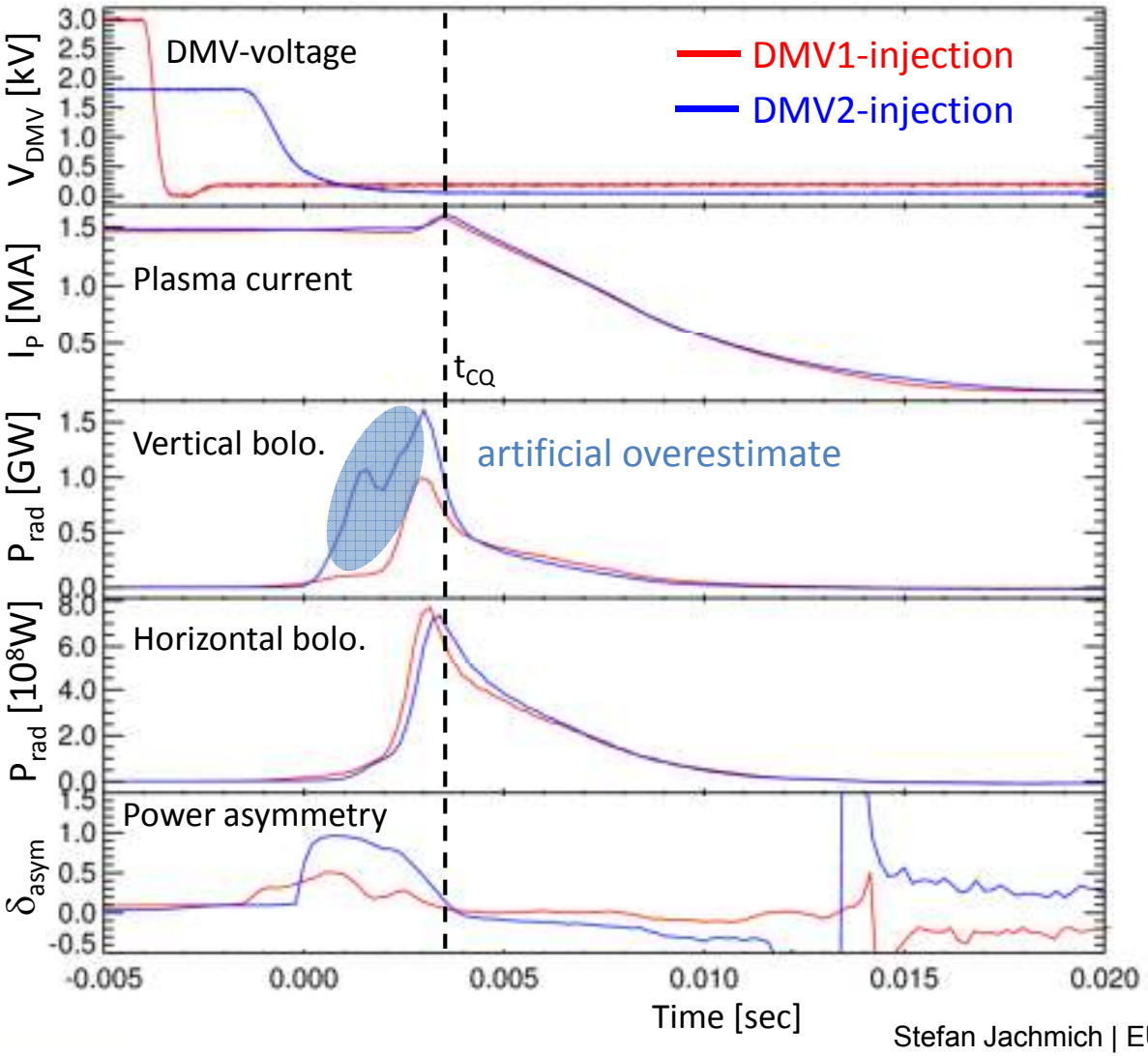
- Horizontal bolometer agrees well with radiated energy determined from combined bolometer measurement if (fudge) factor of 1.25 is applied.

Note: if not stated otherwise, W_{rad} is derived from horizontal bolometer channels.



Comparison DMV1 with DMV2

- Injection of 9.7 barL D_2+10%Ar, W_{mag}~5.6MJ, $W_{thermal}$~1.9MJ, $W_{coupled}$~3.0MJ



Vessel force:

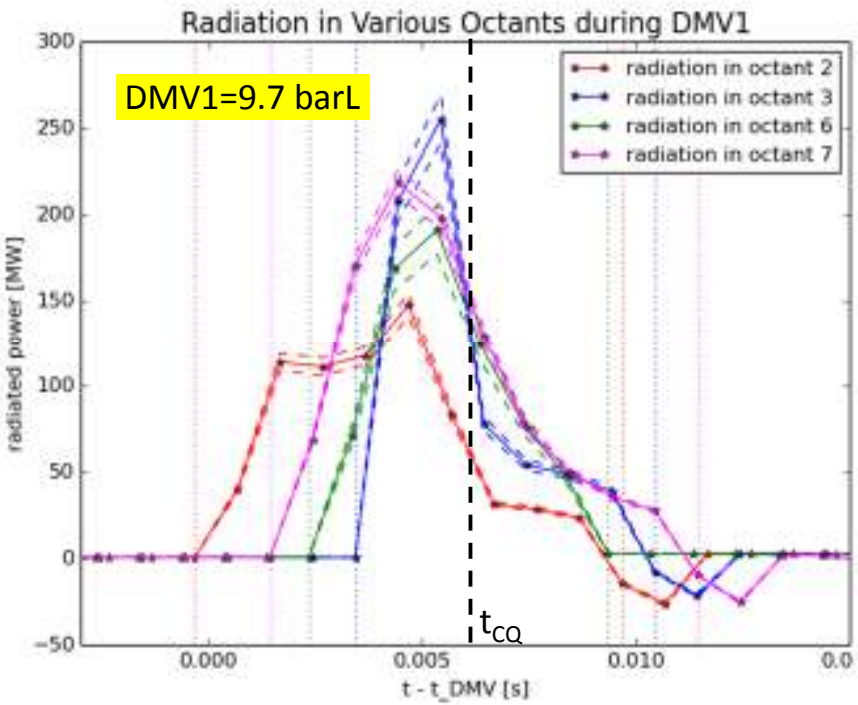
- DMV1: ~75kN
- DMV2: ~74kN

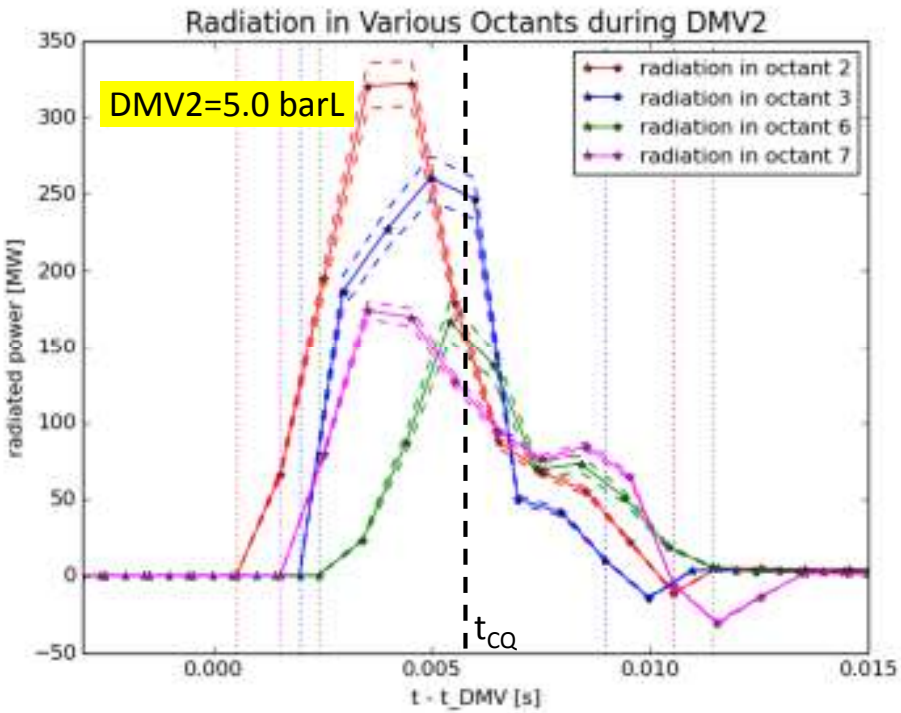
Radiated energy:

- DMV1: 2.50MJ
- DMV2: 2.45MJ

Toroidal asymmetries:

- KB1-measurements in 4 Octants on shot-to-shot basis.
- Data were cross-calibrated with KB5.
- Radiation timing: DMV1: Oct2->Oct7->Oct 3&6, DMV2: Oct2->Oct3->Oct7->Oct6
- Toroidal asymmetries larger for DMV2
- Toroidal radiation peaking factor: DMV1 0.9 to 1.3; DMV2 0.6 to 1.4



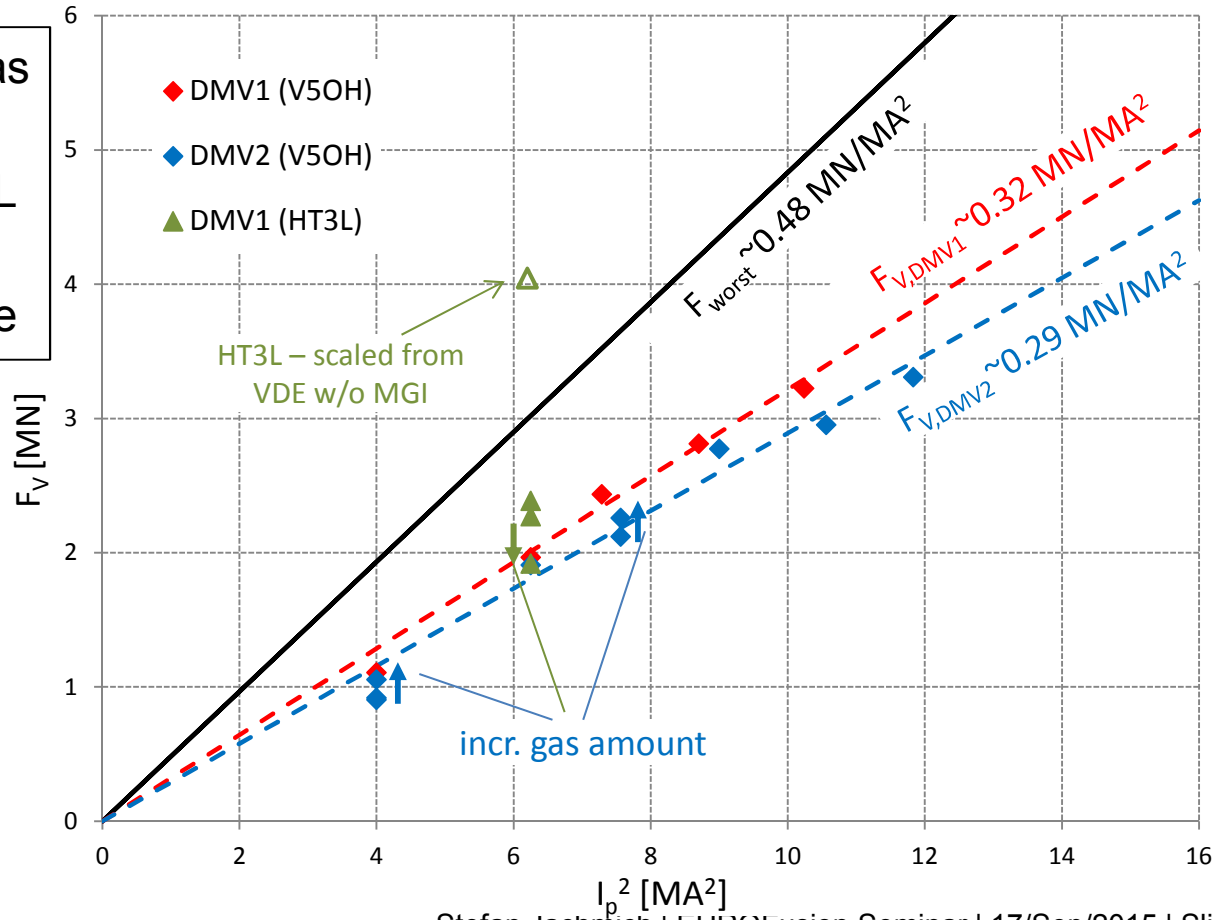


P. Drewelow et al., EPS 2015

Vessel force scaling

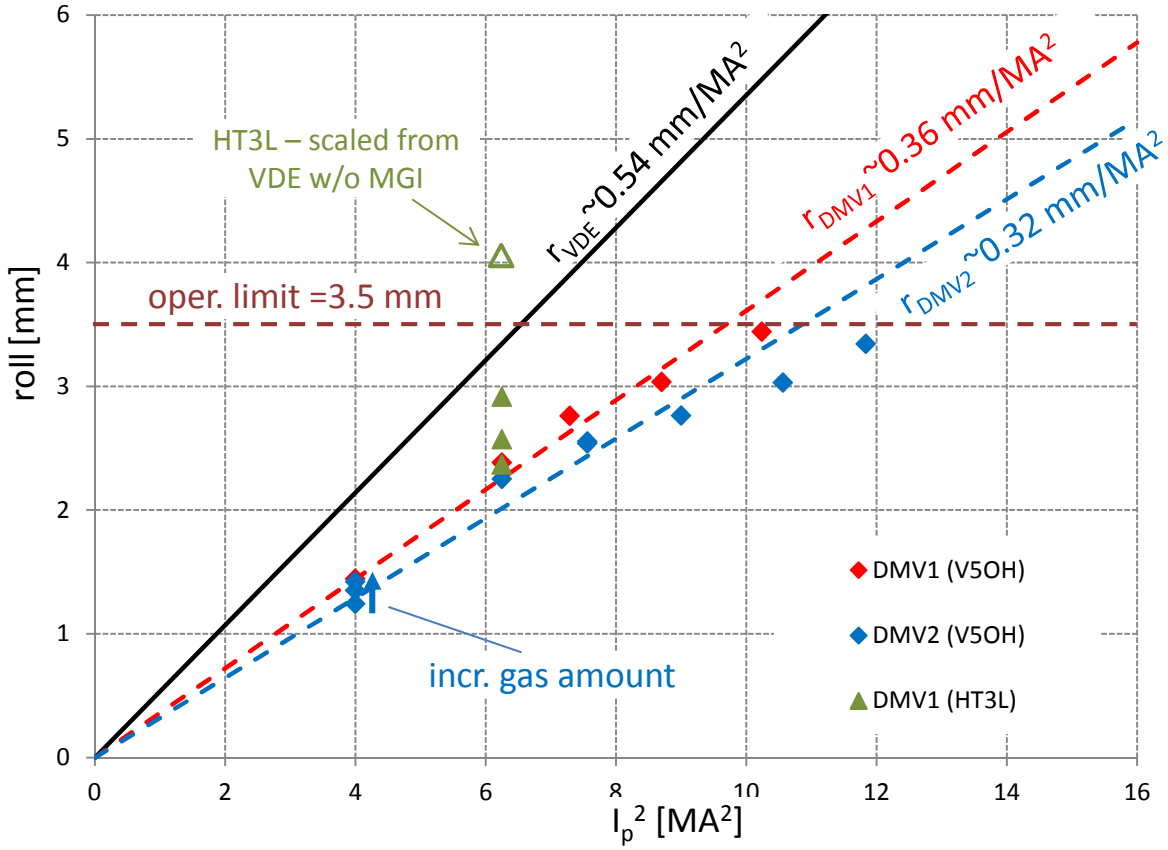
- Dynamic vessel forces are reduced by about 33% (DMV1) and 40% (DMV2).
- No reduction in efficiency has been observed at high Ip.
- Limit of 650tonnes would be achieved at 4.7MA (unmitigated at 3.7MA).
- Smaller gas amount helps reducing forces further.
- High-δ: Fv reduced by 44%-53% (2.0 and 8 barL) at similar level like low-δ.

- MGI into ohmic plasmas
- DMV1: 10% Ar, 7 barL
- DMV2: 10% Ar, 24 barL
- Scaling based on constant DMV-pressure



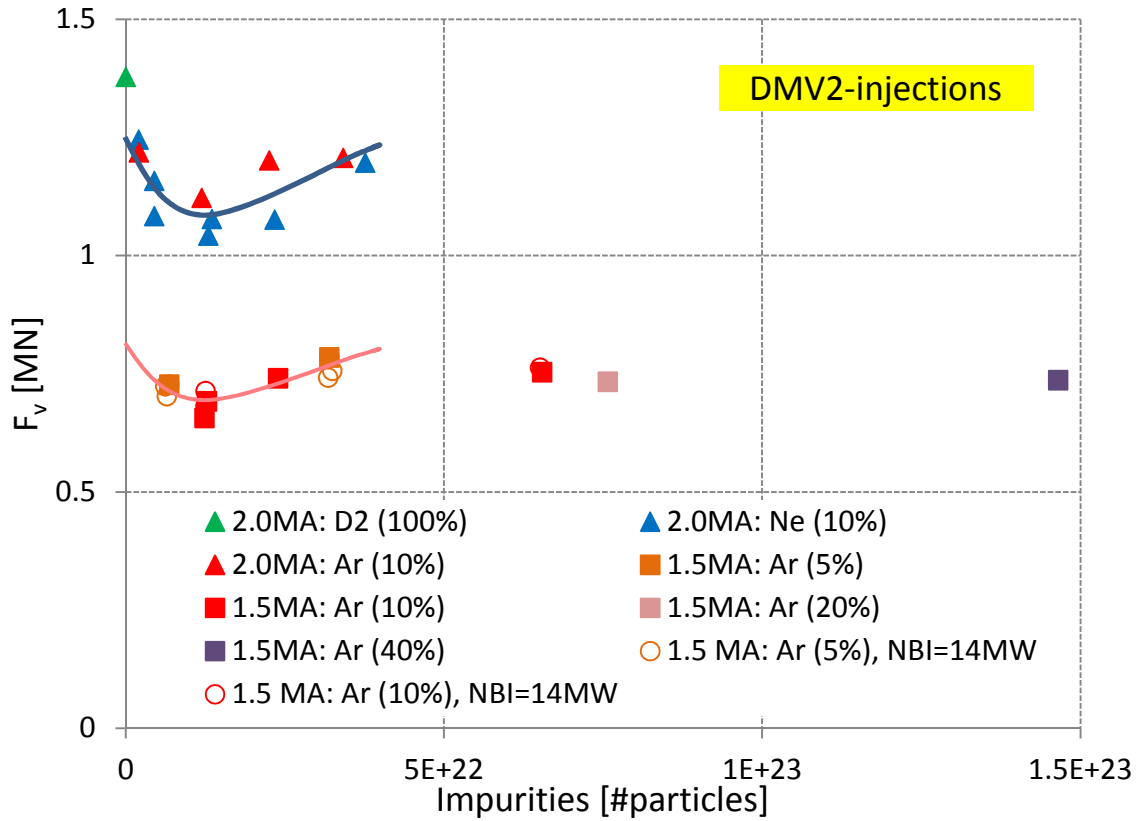
Vessel rolling motion

- Rolling motion is linked to vessel force.
- Operational limit of 3.5 mm is already reached at 3.3MA (unmitigated at 2.5MA).



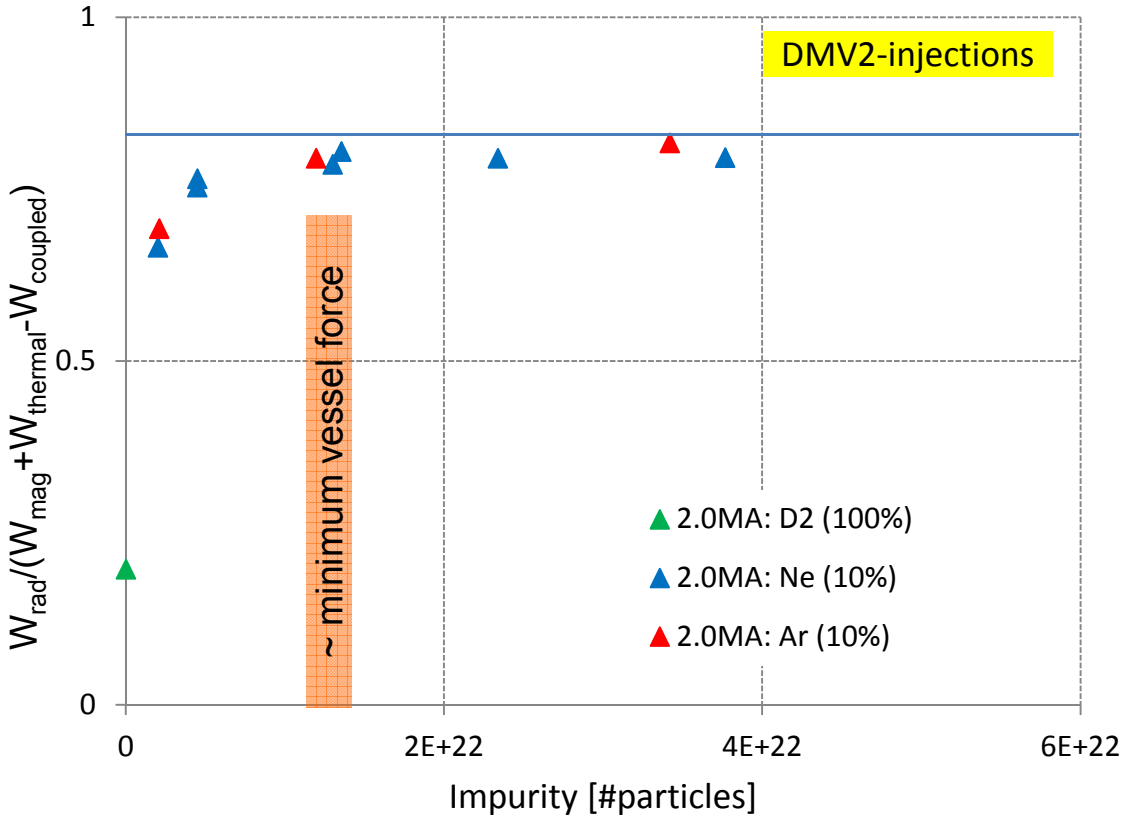
Optimising vessel force reduction

- Data at 2.0MA disruptions suggest minimum of vessel force at low gas amount.
- Higher Ar-concentration does not lead to further reduction of vessel forces.



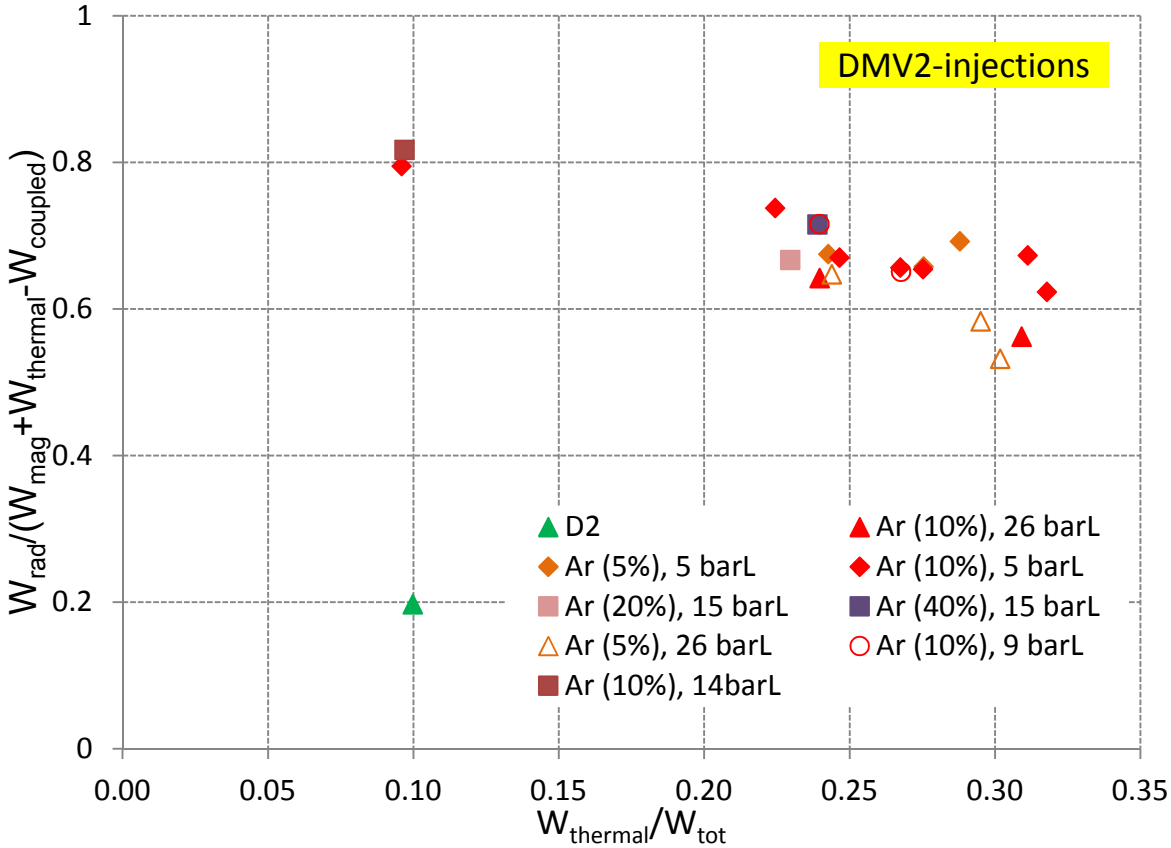
Optimum for radiation efficiency

- Radiated fraction does not further increase with increasing MGI-impurities
- However, required minimum injection might depend on thermal energy
- Note: uncertainty in radiated energy due to toroidal asymmetries



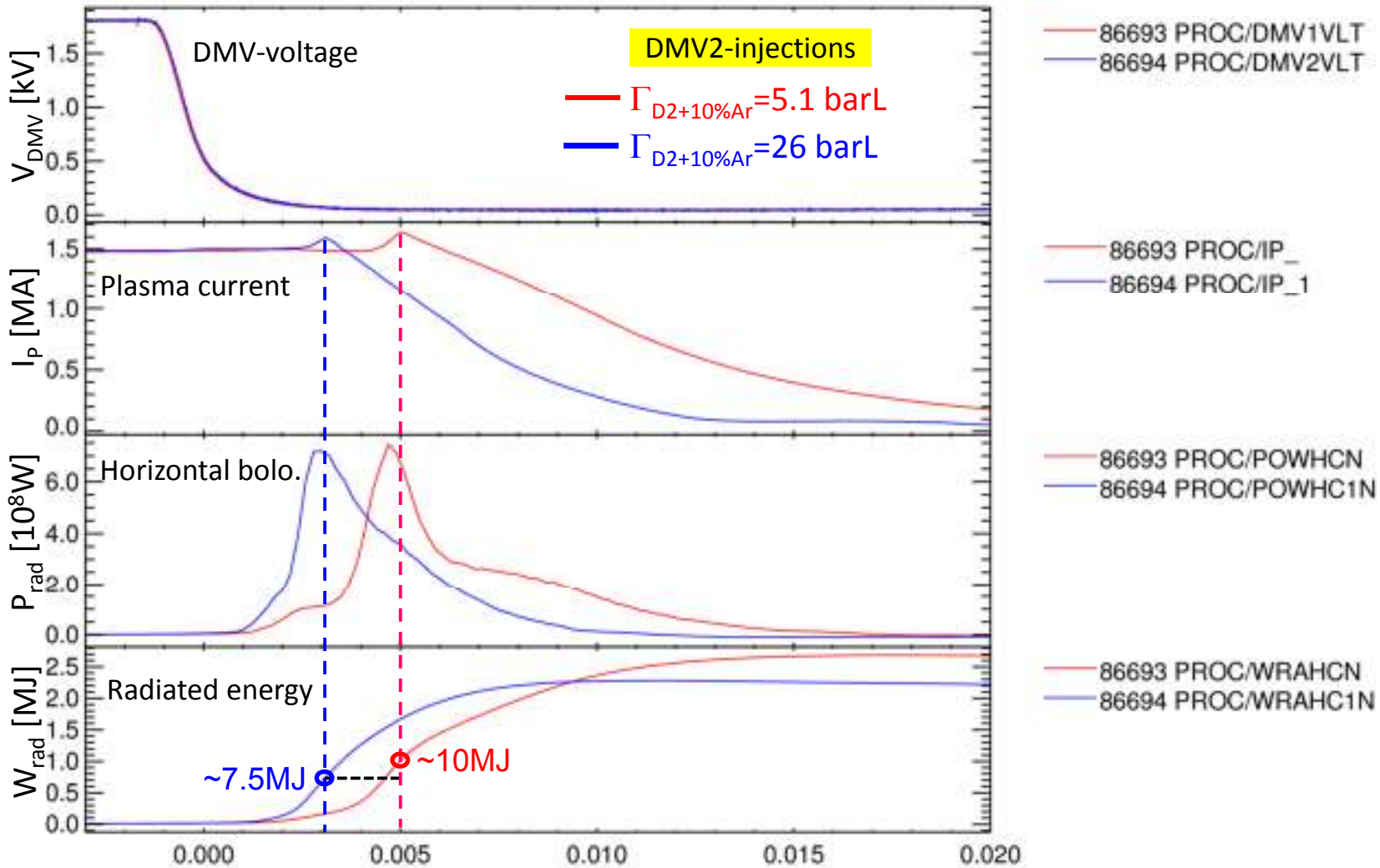
Mitigation efficiency for high thermal energy

- Higher gas amount does not result in higher radiated power fraction
- Minimum required injection cannot be determined due to insufficient data at low injection



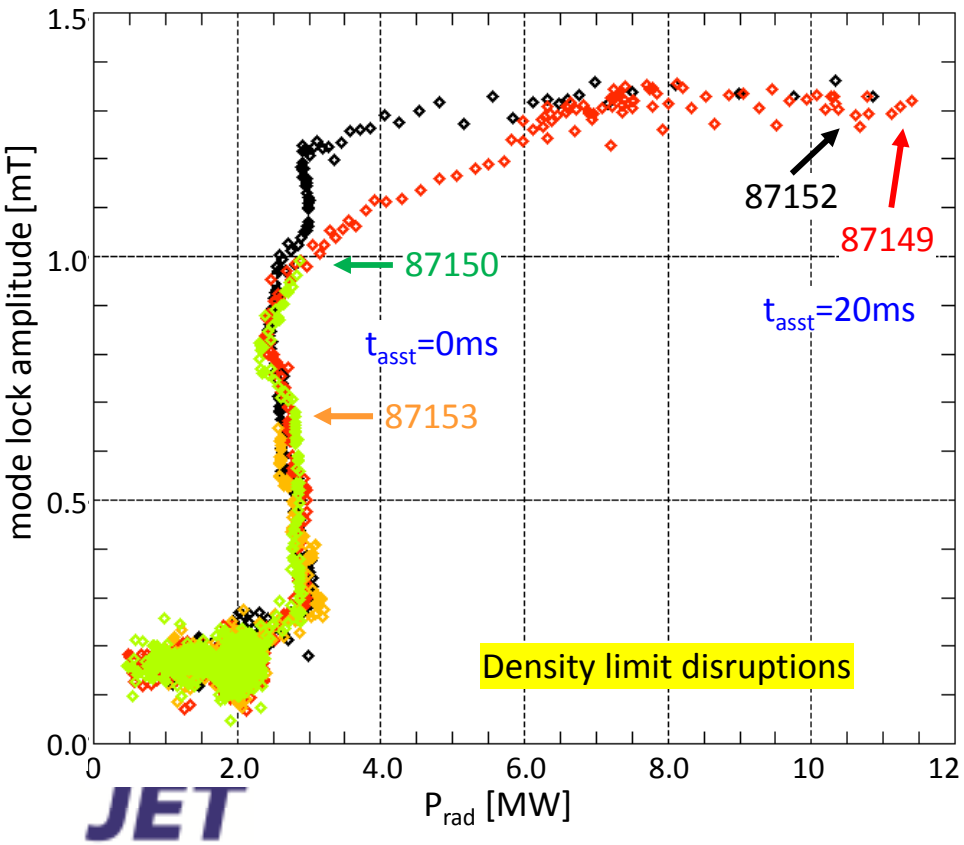
DMV2 – pressure scan

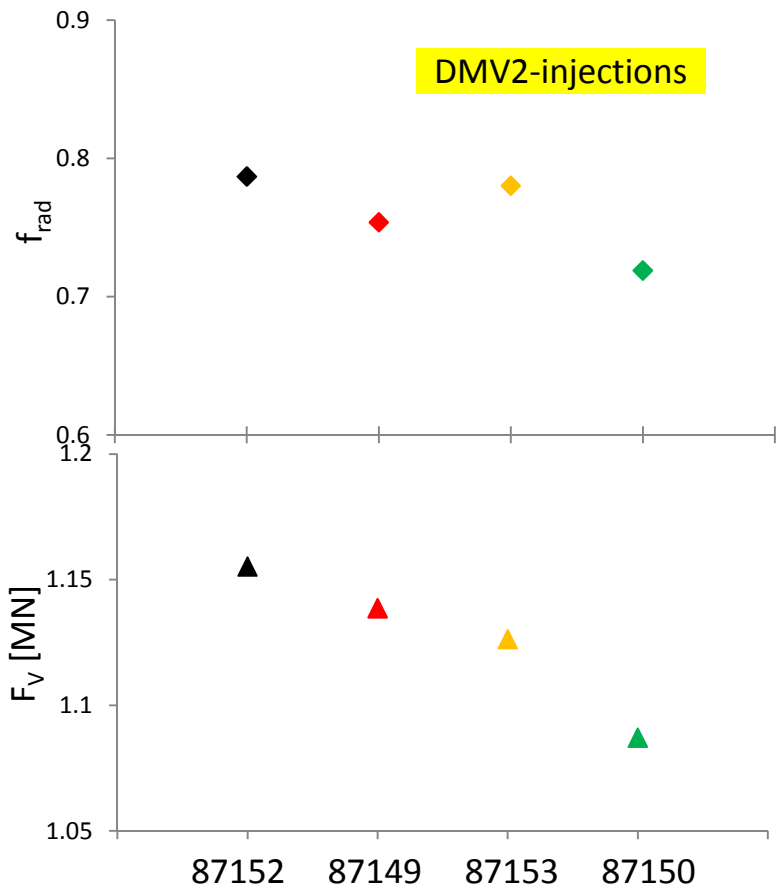
- Toroidal symmetry is achieved faster for higher gas pressure
- Shorter t_{TQ} leads to lower radiated energy before CQ



Optimising disruption detector

- Various settings (level and assertion time) to detect locked mode in density limit pulses have been tested
- Vessel force can be further reduced, minor effect on radiation efficiency





Summary

- Poloidal midplane massive gas injection does not lead to less efficient disruption mitigation.
- Optimum injection rate for disruption force mitigation is in the low DMV2-pressure range: ~5 barL (D_2+10%Ar).

  Note: presently 10 barL (D_2+10%Ar) is used for mitigation in standard ops.
- Mitigation efficiency does not benefit (c.f. f_{rad}) from higher injection rate.
- Disruption detector can be further optimised.
- Higher DMV-pressure helps reducing toroidal asymmetries in TQ-phase.
  - However, too many particles are injected causing shorter TQ-phase.
  - Reduce DMV2-volume by inserting filler?
  - Test highest possible DMV2-pressure (50bar) in session without divertor cryo?
- Finding the optimum MGI-configuration remains a challenge.

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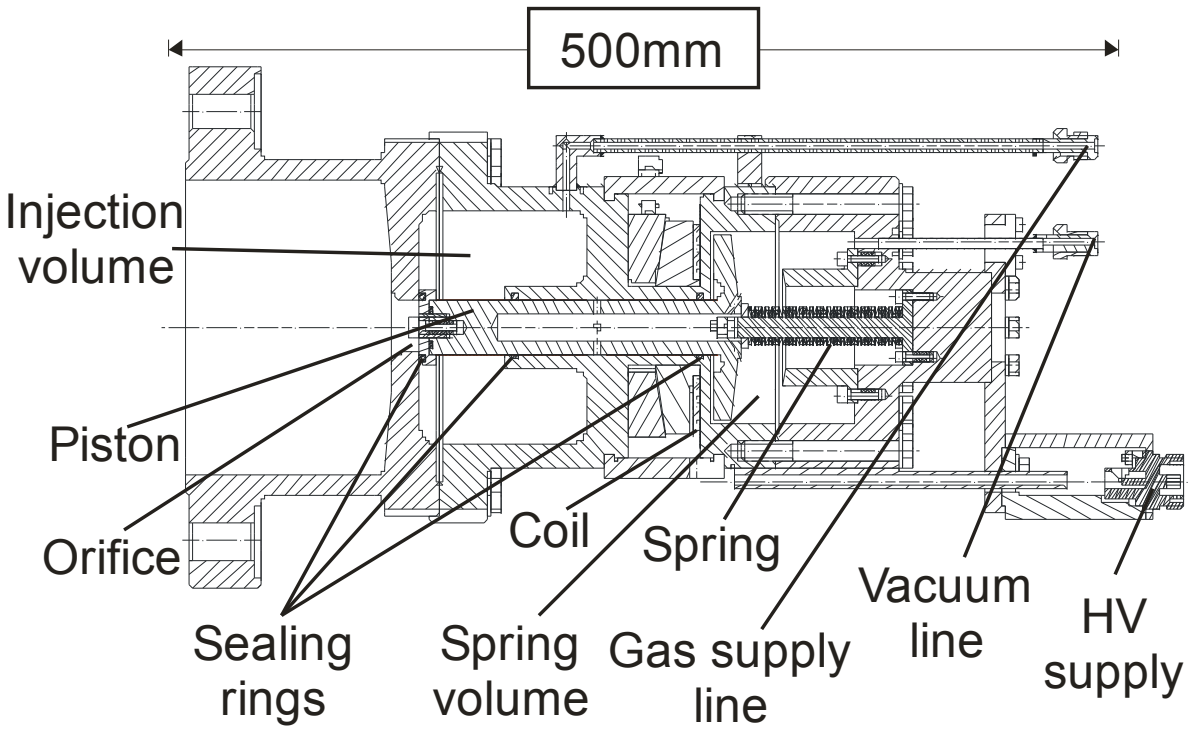
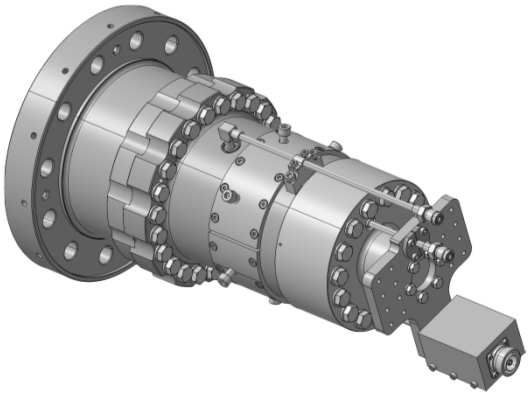
Thank you

Massive gas injection valve





- **Fast valve specifically designed for tokamak environments**
  Eddy current driven valve
  insensitive to magnetic fields
  (up to 2 T tested in TEXTOR on “DMV3”)

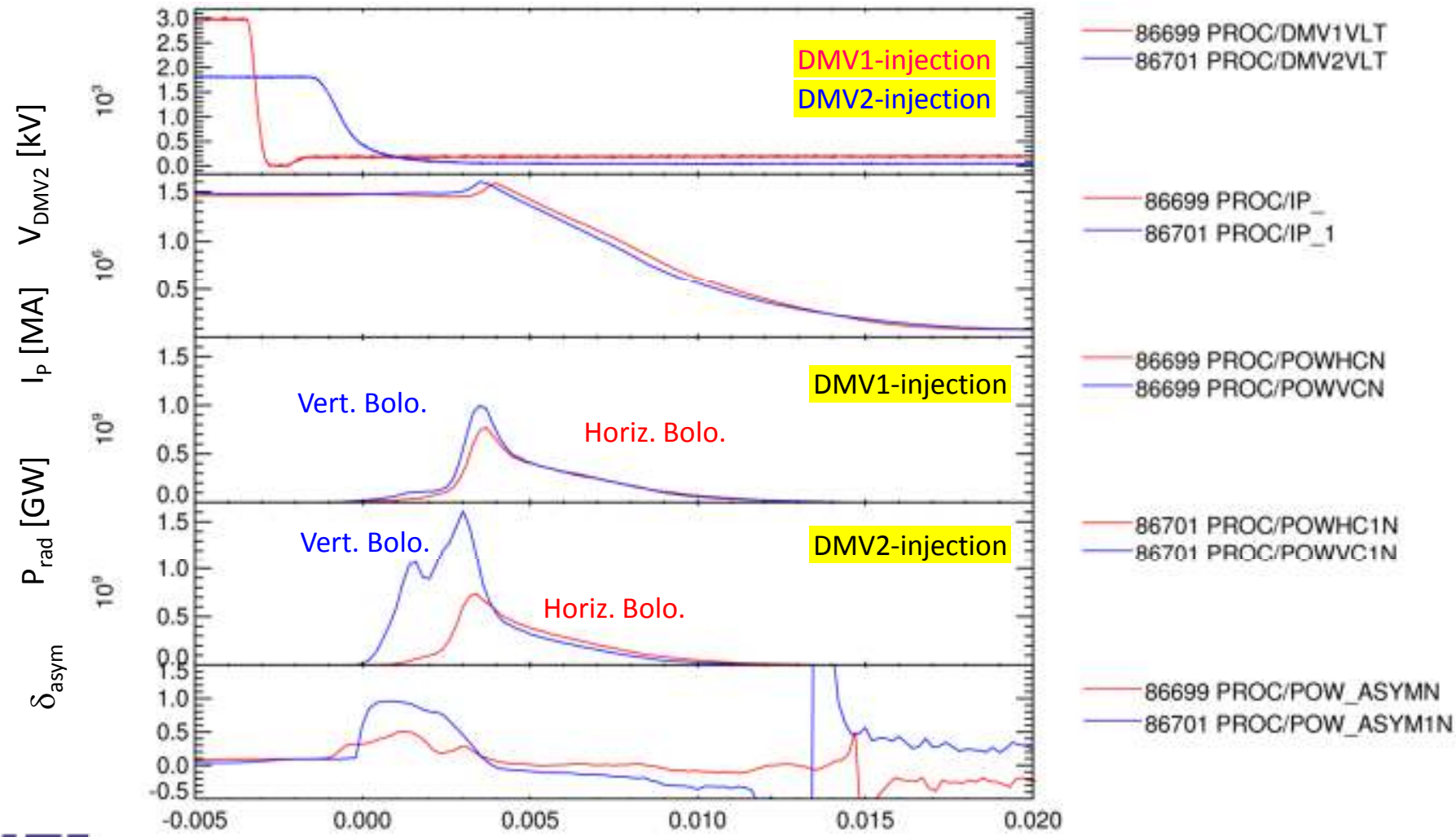
- **All metal valve***
  *O-ring Kalrez seal towards vacuum and radial seal between injection and spring volume
  No electrical feed throughs – external coil

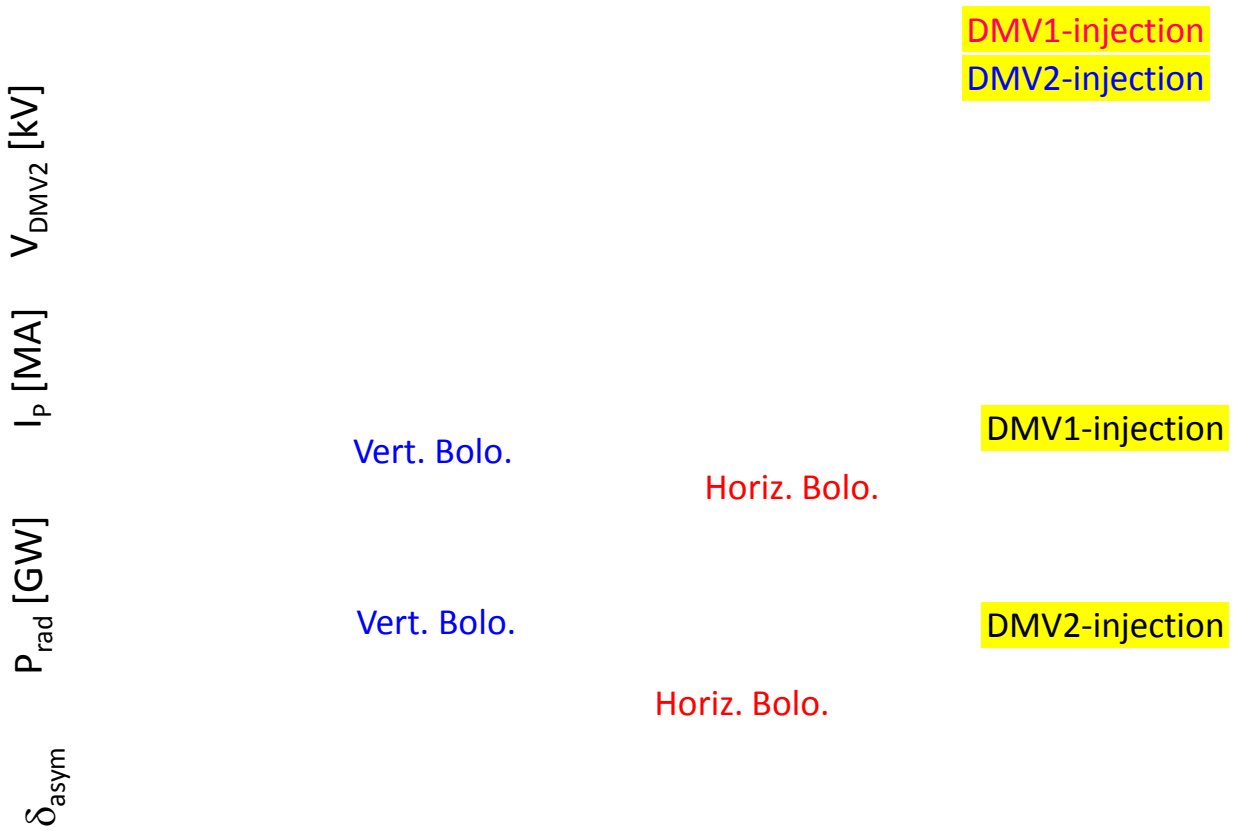
- **Reliable design**
  Pressure tested up to 8.0MPa – 5.0 MPa operational pressure

JET

Comparison DMV1 with DMV2

- Data at



Dummy slide

- Data at